



Roll No:

BPHARM
(SEM. I) THEORY EXAMINATION 2025-26
PHARMACEUTIAL ANALYSIS-1

TIME: 3 HRS**M.MARKS: 75**

Note: Attempt all Sections. In case of any missing data, choose suitably.

SECTION A

1. Attempt all questions in brief.

10 x 2 = 20

a.	What is meant by diazotization?
b.	How will you estimate calcium gluconate?
c.	Outline the significance of modified Volhard's method.
d.	What do you mean by redox reaction?
e.	Differentiate between primary and secondary standards.
f.	Explain the concepts of molarity and molality.
g.	State the principle underlying polarography.
h.	Classify the electrodes in potentiometric titration.
i.	List the different solvents employed in non-aqueous titrations.
j.	Distinguish between accuracy and precision.

SECTION B

2. Attempt any two parts of the following:

2 x 10 = 20

a.	Classify acid-base indicators and explain the underlying theory governing their action, supported by suitable examples.
b.	Explain the principle of complexometric titrations. Describe the role of EDTA as a titrant and discuss the indicators used with suitable examples.
c.	Describe the various types of errors encountered in pharmaceutical analysis and discuss the approaches used to reduce or eliminate them.

SECTION C

3. Attempt any five parts of the following:

7 x 5 = 35

a.	Explain the fundamental principles, instrumentation, and practical applications of conductometric analysis.
b.	Differentiate between iodimetric and iodometric titrations with suitable examples.
c.	Discuss the various sources from which impurities may arise in pharmaceutical substances.
d.	Describe the different analytical techniques employed in pharmaceutical analysis.
e.	Explain the principle of gravimetric analysis and outline the procedural steps involved.
f.	Explain the construction, working principle, and applications of a glass electrode for pH measurement. Add a neat labeled diagram.
g.	Explain the various neutralization curves of acid-base titration.